

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 6: Solving Problems Based on Quadratic Equations

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Problems on speed, distance and time

1. A couple drive 350 km to a hill station. Heavy rain reduces their average speed by 20 km/h, and the journey takes 2 hours longer. Find (a) the original speed of the car (b) the time taken at the reduced speed. **ICSE 2025, 4 marks**

SOLUTION

(a) Let the original speed be x km/h. $\frac{350}{x-20} - \frac{350}{x} = 2$
 $350x - 350(x-20) = 2x(x-20)$, so $7000 = 2x^2 - 40x$ and $x^2 - 20x - 3500 = 0$
 $(x-70)(x+50) = 0$, so $x = 70$ (speed cannot be negative).
 (b) Reduced speed = 50 km/h, so the time is $\frac{350}{50} = 7$ hours.

Answer: (a) 70 km/h (b) 7 hours

2. A man travels 100 km at a uniform speed of x km/h. At a speed 5 km/h faster, the journey would have taken 1 hour less. Find x . **ICSE 2023, 3 marks**

SOLUTION

$\frac{100}{x} - \frac{100}{x+5} = 1$
 $100(x+5) - 100x = x(x+5)$, so $x^2 + 5x - 500 = 0$
 $(x+25)(x-20) = 0$, so $x = 20$ (speed cannot be negative).

Answer: 20 km/h

2 Problems on numbers

3. Two natural numbers differ by 7 and their product is 450. Find the numbers. **ICSE 2020, 3 marks**

SOLUTION

Let the smaller number be x . Then $x(x+7) = 450$, so $x^2 + 7x - 450 = 0$.
 $(x+25)(x-18) = 0$, so $x = 18$ (a natural number cannot be negative).

Answer: 18 and 25

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.

MathSetu